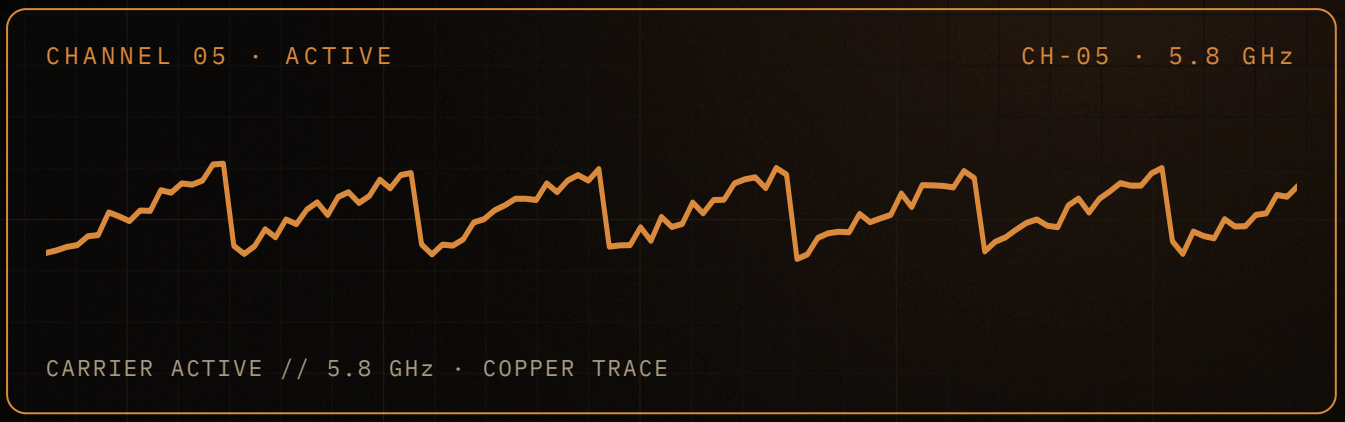


CLASSIFIED — PROBLEM STATEMENT CHANNEL

INDUSTRIAL AUTOMATION & MONITORING

PROBLEM STATEMENTS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29-30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 05 CH-05 · 5.8 GHz



CARRIER ACTIVE // 5.8 GHz · COPPER TRACE

INDUSTRIAL AUTOMATION & MONITORING

The demands of modern manufacturing, assembly floors and heavy-machinery uptime through predictive intelligence. Ruggedized data acquisition and actuation that preempt costly breakdowns and optimize throughput.

— AREAS TO EXPLORE

- Predictive Maintenance
- IIoT Gateways
- Machine Vision Quality
- Automated Sorting
- PLC & SCADA Interfacing

— WHO SHOULD EXPLORE

Instrumentation · Mechatronics · Power Electronics · Sensor Fusion & Predictive Telemetry

> *"The heartbeat of heavy industry — engineer the feedback loops that keep the factories running."*

— IN THIS CHANNEL

PS-01	Early Detection of Conveyor Belt Jamming & Material Spillage
PS-02	Compressed Air Leakage Detection & Energy Loss Estimation
PS-03	Early Detection of Industrial Motor Bearing Failure Using Vibration Analysis
PS-04	Monitoring & Identification of Unexpected Industrial Machine Downtime
PS-05	Drone-Based Inspection of Industrial Structures & Hard-to-Reach Equipment

PS-01 EARLY DETECTION OF CONVEYOR BELT JAMMING & MATERIAL SPILLAGE

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Belt conveyors move bulk material (coal, cement, grain, packaged goods) through plants and mines. A jam at a transfer point, a stalled or slipping belt, or material falling off the belt can stop production, damage the belt and create safety hazards. The challenge is to detect these conditions early using simple sensing and to stop or alert before damage or large spillage occurs.

B · CONTEXT

Conveyor stoppages are often discovered by operators walking the line, sometimes only after the motor has tripped or material has piled up. Belt slip and blockages waste energy, wear the belt and can cause fires from friction. Spilled material demands manual clean-up and creates slip and dust hazards. A low-cost monitoring and interlock system that recognises abnormal behaviour within seconds would reduce downtime, damage and risk to workers.

C · CORE FEATURES

- Monitor motor current and belt speed to detect jamming, overload and belt slip.
- Detect material build-up at chutes and transfer points using proximity, level or optical sensing.
- Detect material spillage or belt misalignment where practical.
- Raise alarms and trigger a controlled stop, interlocked with upstream feeders.
- Log events with time stamps for maintenance review.

D · DESIGN REQUIREMENTS

- Users: mines, cement and fertiliser plants, ports, food processing and warehouses.
- Cost: retrofittable on existing conveyors without long shutdowns, affordable for small and medium plants.
- Robustness: tolerates dust, vibration, moisture and wide temperature variation; low maintenance.
- Safety: supports safe stopping and lock-out practice; false stops should be rare as they also cost production.

E · TECHNICAL FOCUS

SENSING Motor current (CT or Hall sensor), belt speed (encoder or inductive/Hall pulse sensor) compared with expected speed, and chute-level detection (inductive, IR, ultrasonic or capacitive).

PROCESSING & CONTROL Microcontroller or PLC-style logic with thresholds, time delays and start-up inhibit; slip is typically evaluated as the ratio of measured to expected belt speed.

COMMUNICATION Local indicator and relay output to the motor starter; optional wired or wireless link to a supervisory dashboard.

POWER SUPPLY Industrial 24 V DC or isolated supply from the mains.

SIMULATION Conveyor and motor load model in MATLAB/Simulink, Python or equivalent, showing normal running, overload, jam and slip events and the detection logic response.

OPTIONAL Camera-based spillage detection, belt-edge misalignment sensing, predictive trending, integration with SCADA.

F · EXPECTED OUTCOME

- System architecture showing sensor positions and interlock logic.
- Simulation of normal operation and at least three abnormal events, with detection time for each.
- Threshold selection and a discussion of false-alarm handling during start-up and load changes.
- Estimated cost and installation approach for retrofit on an existing conveyor.
- Expected benefits in reduced downtime and damage.

PS-02 COMPRESSED AIR LEAKAGE DETECTION & ENERGY LOSS ESTIMATION

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Compressed air is one of the most expensive utilities in a factory because only a small fraction of the electrical energy used by a compressor ends up as useful pneumatic work. Leaks at joints, hoses, valves and fittings waste air continuously, even when the plant is idle. The challenge is to detect and quantify leakage and to estimate the resulting energy loss and cost.

B · CONTEXT

In poorly maintained plants, leakage commonly consumes a significant share (often 20–30%) of compressor output. Leaks are hard to hear in noisy workshops and are usually ignored until pressure drops. Manual surveys with ultrasonic detectors are periodic and labour-intensive. A monitoring approach that measures leakage rate, shows how much energy and money is lost, and prioritises where to look would help plants cut energy use and reduce compressor wear.

C · CORE FEATURES

- Measure system pressure and, where possible, air flow or compressor loading over time.
- Estimate the leakage rate during non-production periods when demand should be near zero.
- Convert estimated leakage flow into energy loss and monetary cost using a compressor specific-energy assumption.
- Alert when the leakage fraction exceeds a set limit, and track trends after repairs.
- Optionally localise leaks by section or by acoustic detection.

D · DESIGN REQUIREMENTS

- Users: small and medium manufacturers, workshops and process plants.
- Payback: should pay back quickly through energy saved; monitoring cost small versus avoidable losses.
- Deployment: non-intrusive installation without long plant shutdowns, with minimal operator training.
- Sustainability: lower electricity consumption and emissions, fewer compressors running unnecessarily.

E · TECHNICAL FOCUS

SENSING Line pressure sensor(s); optional flow sensor, compressor motor current sensing, or ultrasonic microphone (leak sound is mainly in the tens-of-kHz range).

ANALYTICS Pressure-decay testing with supply isolated, or load/unload-time analysis of the compressor with no demand. State the energy assumption (for 7 bar rotary-screw compressors, specific power is often 6–8 kW per m^3/min of free air).

PROCESSING & COMMUNICATION Microcontroller with local display, and optional Wi-Fi or industrial communication to a dashboard.

POWER SUPPLY Industrial 24 V DC or mains-derived supply.

SIMULATION Receiver tank and compressor model with leak orifices of different sizes in MATLAB/Simulink, Python or equivalent; show how pressure and compressor duty change with leakage.

OPTIONAL Ultrasonic leak-localisation handheld, flow-meter integration, automated weekend leak report.

F · EXPECTED OUTCOME

- System architecture and measurement plan.
- Simulation showing pressure behaviour and compressor duty for no-leak and at least three leak sizes.
- Leakage estimate (L/min or m^3/min) with derived energy loss (kWh/year) and cost, with assumptions stated.
- Estimated accuracy and limitations of the method.
- Cost-benefit estimate for a representative plant.

PS-03 EARLY DETECTION OF INDUSTRIAL MOTOR BEARING FAILURE USING VIBRATION ANALYSIS

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Rolling-element bearings are among the most common causes of motor failure. As a defect (a crack or spall on the outer race, inner race or a rolling element) develops, it produces tiny repetitive impacts that change the vibration signature long before the motor stops. The challenge is to detect these changes early from vibration measurements and to estimate the health of the bearing.

B · CONTEXT

Unexpected bearing failure causes unplanned shutdowns, secondary damage and safety risk, while routine replacement wastes serviceable parts. Handheld vibration checks are infrequent and depend on operator skill. Simple overall vibration levels only rise late in the failure process. Continuous, low-cost vibration monitoring with signal processing that reveals early defect signatures would allow maintenance to be planned and avoid costly breakdowns.

C · CORE FEATURES

- Acquire vibration signals from the motor bearing housings with sufficient bandwidth.
- Extract condition indicators such as RMS, kurtosis, crest factor and frequency-domain features.
- Detect bearing defect frequencies using spectrum and envelope analysis.
- Compute a health index and trend it over time; issue graded alerts.
- Display trends and store data for maintenance planning.

D · DESIGN REQUIREMENTS

- Users: manufacturing plants, pumping stations, HVAC and utility facilities with many motors.
- Cost: monitoring cost per motor must be low relative to the cost of a failure; use affordable MEMS sensors.
- Maintenance value: earlier warning with fewer false alarms, enabling planned rather than emergency repair.
- Practicality: easy mounting, tolerant of industrial noise and temperature, aligned with vibration-severity guidance (e.g. ISO 20816).

E · TECHNICAL FOCUS

SENSING Accelerometer (MEMS or piezoelectric) with bandwidth of several kHz and a stated sampling rate; discuss mounting location and axis.

SIGNAL PROCESSING Filtering, FFT, envelope (demodulation) analysis and statistical indicators. Defect frequencies (BPFO, BPFI, BSF, FTF) follow from bearing geometry and shaft speed; show how they are used.

PROCESSING PLATFORM Microcontroller, single-board computer or edge device; discuss whether analysis runs on the sensor node or a gateway.

DATA Public bearing datasets (e.g. CWRU, Paderborn, NASA IMS) or synthetic signals with impulse trains, resonance and noise.

SIMULATION MATLAB, Python or equivalent to generate or load vibration data for healthy and faulty bearings and demonstrate detection.

OPTIONAL ML classification, remaining-useful-life estimation, temperature and motor-current fusion, wireless sensor network.

F · EXPECTED OUTCOME

- Monitoring architecture and sensor mounting plan.
- Signal-processing chain with spectra and envelope spectra for healthy and faulty cases.
- Detection results for at least three fault types or severities, with a health-index plot.
- Evaluation of detection sensitivity and robustness to noise.
- Cost estimate and a scalable approach for a plant with many motors.

PS-04 MONITORING & IDENTIFICATION OF UNEXPECTED INDUSTRIAL MACHINE DOWNTIME

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Machines stop for many reasons: power interruption, material shortage, jams, tool changes, overloads, faults or operator breaks. In many plants these stoppages are recorded by hand, if at all, with vague causes. The challenge is to detect machine state automatically (running, idle, stopped, faulted), time-stamp each unexpected stoppage and identify its likely cause from available signals.

B · CONTEXT

Unplanned downtime directly cuts output and delivery reliability, but managers cannot reduce what they cannot measure. Manual logs are incomplete and subjective, so improvement efforts target the wrong problems. A low-cost monitoring system that detects stops, classifies likely causes and summarises downtime statistics would let plants focus on the biggest losses, improve maintenance planning and raise overall equipment effectiveness without costly machine upgrades.

C · CORE FEATURES

- Detect machine state from signals such as motor current, power, vibration, cycle count and status contacts.
- Record start and end time of every stoppage and separate planned from unexpected events.
- Classify the probable cause (power loss, material starvation, overload, jam or operator stop) using rules or learned patterns.
- Compute availability, MTBF and MTTR, and rank causes by lost time (Pareto).
- Provide a live status view and automatic shift or daily reports.

D · DESIGN REQUIREMENTS

- Users: small and medium manufacturers with older machines that lack built-in data logging.
- Cost & installation: non-intrusive sensors, no changes to machine control logic, quick installation.
- Value: actionable downtime reports that guide maintenance and process improvements.
- Data handling: reliable logging during network outages, secure storage and a simple operator interface.

E · TECHNICAL FOCUS

SENSING Clamp-on current sensors, vibration or sound sensors, proximity/cycle sensors and optional status-light or contact monitoring; choose the minimum set that separates the main states.

PROCESSING & LOGIC Microcontroller or edge device running a state machine with debouncing and minimum-duration rules; cause classification by thresholds, decision rules or a simple model.

COMMUNICATION Wi-Fi, Ethernet, RS-485/Modbus or LoRa gateway; local buffering when the network is down.

DATA & DASHBOARD Time-series storage and charts of state timeline, downtime Pareto and key metrics.

SIMULATION Synthetic multi-day machine data containing several kinds of stoppage, created in Python or equivalent; report classification accuracy.

OPTIONAL Operator reason-code entry, predictive alerts from repeated micro-stops, multi-machine line-balancing view, ERP/MES integration.

F · EXPECTED OUTCOME

- System architecture and the signals chosen for each machine type.
- State-detection and cause-classification logic with a flowchart.
- Simulation over several simulated shifts, with detection accuracy (%) and classification accuracy (%).
- Sample reports: downtime timeline, Pareto chart, MTBF/MTTR values.
- Cost estimate and a plan for scaling from one machine to a whole shop floor.

PS-05 DRONE-BASED INSPECTION OF INDUSTRIAL STRUCTURES & HARD-TO-REACH EQUIPMENT

Recommended for Third & Final Year students

— A · PROBLEM DEFINITION

Chimneys, boilers, storage tanks, cooling towers, pipe racks, roofs and transmission structures are costly and risky to inspect because they require scaffolding, rope access or shutdowns. The challenge is to design an unmanned aerial vehicle (UAV) based inspection system that plans safe flight around such structures, captures useful inspection data and converts it into georeferenced findings, while coping with confined spaces, metal structures and limited flight time.

— B · CONTEXT

Drones can reduce inspection time, cost and human exposure, but industrial sites are demanding: GPS may be weak near tall metal structures, wind and heat disturb flight, flight time is limited and image quality varies with lighting. Raw footage has little value unless defects such as corrosion, cracks, leaks and hot spots are identified and located. A well-designed system links mission planning, payload, safe flight and data analysis into a repeatable workflow.

— C · CORE FEATURES

- Plan an inspection route that covers the target structure with defined image overlap, distance and viewing angle.
- Carry a suitable payload (visual camera, thermal camera, optional ultrasonic thickness or gas sensor).
- Support safe flight through obstacle awareness, geofencing, return-to-home and fail-safe behaviour.
- Detect and label defects (corrosion, cracks, hot spots, leaks) in the captured data and tag them with location.
- Generate an inspection report with annotated images and defect severity.

— D · DESIGN REQUIREMENTS

- Users: refineries, power plants, cement and steel plants, utilities and infrastructure owners.
- Value: reduced shutdown time, no scaffolding, lower risk to personnel compared with manual inspection.
- Safety & regulation: consideration of Indian drone regulations, site permission, hazardous-area limits and operator training.
- Cost & repeatability: affordable platform choices and the ability to repeat a mission to compare inspections over time.

— E · TECHNICAL FOCUS

PLATFORM Multirotor UAV (or tethered/caged variant for confined spaces); compute a payload and endurance budget (battery capacity, hover power, mass) rather than assuming a product.

NAVIGATION & SENSING GNSS, IMU and barometer, plus cameras, LiDAR, ultrasonic or optical-flow sensing for GPS-poor areas; discuss magnetic interference near steel.

MISSION & DATA PROCESSING Waypoint and coverage planning, image-quality checks and defect detection using classical image processing or ML on sample images.

COMMUNICATION Telemetry and video link with range and latency considered; failsafe on link loss.

SIMULATION Flight-path coverage and energy simulation in MATLAB, Python or a UAV simulator (e.g. PX4 SITL with Gazebo) combined with defect detection on sample images.

OPTIONAL Thermal and visual image fusion, automatic 3D reconstruction, tethered power for longer endurance, digital-twin comparison between inspections.

— F · EXPECTED OUTCOME

- System architecture covering platform, sensors, communication and ground software.
- Simulated inspection mission on a representative structure showing path, coverage (%), and battery use.
- Defect-detection demonstration on sample images with quantified detection performance.
- Risk assessment and safety procedures, including behaviour on GPS or link loss.
- Cost estimate and a comparison with conventional scaffold or rope-access inspection.